

Histogram-Based Skin Color Detection Report

Training Data

Training data was sourced from a subset of the 11k Hands dataset, using 100 images. Each image was processed in RGB to extract candidate skin pixels using a set of threshold conditions to filter out pixels that are too dark, too gray, or lack the warm bias characteristic of skin tone. The pixels that pass are then converted to the target color space for histogram construction.

Color Space Selection

Three color spaces were evaluated: H-S (hue-saturation), normalized RGB (nRGB), and a Gaussian model over H-S:

- Normalized RGB divides each channel by the sum of the three channels, reducing the effect of brightness on the color representation. The resulting nR and nG values are used as the two histogram axes.
- H-S uses the hue and saturation channels from the HSV representation. Hue captures the color angle independent of brightness and saturation, and saturation captures color purity independent of value. Together, they form a chrominance-only descriptor that is more compact and more discriminative for skin than nRGB under varied lighting.

Segmentation

At test time, each pixel in the test image is mapped to its corresponding histogram bin, and the stored probability is retrieved. Pixels with probability above a threshold are marked as skin. The threshold is set to 5% of the histogram's peak value, making it adaptive to the training distribution rather than fixed. After thresholding, binary closing with a 9x9 kernel is applied to fill small holes and gaps, followed by binary opening with a 3x3 kernel to remove isolated noise. Connected component labeling (CCL) is then run on the resulting binary mask using 8-connectivity, and the largest component is kept as the final skin region. The output is the original image with all non-skin pixels set to zero.

Results: H-S Histogram and nRGB Histogram

- H-S: On gun1.bmp, the hand is cleanly isolated with all three extended fingers segmented correctly. The jacket sleeve and background are fully excluded. On joy1.bmp, the full hand, including all five fingers, is captured with clean boundaries. The book cover and surrounding objects are not picked up. On pointer1.bmp, the hand and wrist region are correctly segmented. A small gap appears at the ring location, which is expected since the metal band does not satisfy any skin color criterion.

- nRGB: gun1.bmp shows the same three-finger structure with clean background exclusion. joy1.bmp captures the full hand. pointer1.bmp shows the hand and wrist with the same ring gap. This produces slightly harder cutoffs at the wrist and fingertip edges on gun1 and pointer1, which appears to be due to nRGB being more sensitive to the brightness drop at the skin-sleeve boundary. H-S handles these transitions more smoothly because hue and saturation are less affected by the local intensity change.

Results: Gaussian Model (Extra Credit)

A parametric Gaussian model was trained on the H-S pixels using a distance threshold of 2.25. While more memory-efficient than a 64x64 histogram, the Gaussian model assumes a strict elliptical distribution. Its softer decision boundaries resulted in minor false positives on the clothing in joy1.bmp and pointer1.bmp, which the histogram model successfully avoided.